

STAT 140 Section 6 — Solutions: Diet Experiment

Setup. A scientist tests diet A (treatment) vs. diet B (control) on 20 animals ($P = 10$ pairs). Within each pair animals are similar; one receives each diet by coin flip. Response: percentage decrease in body weight after two months.

Pair	1	2	3	4	5	6	7	8	9	10
Diet A (unit)	13.2 (1)	8.2 (2)	10.9 (1)	14.3 (1)	10.7 (2)	6.6 (2)	9.5 (1)	10.8 (2)	8.8 (2)	13.3 (1)
Diet B (unit)	14.0 (2)	8.8 (1)	11.2 (2)	14.2 (2)	11.8 (1)	6.4 (1)	9.8 (2)	11.3 (1)	9.3 (1)	13.6 (2)

(i) What are the drawbacks of a completely randomised treatment assignment?

With 20 animals differing in age, sex, and other characteristics, a CRE assigns animals to diets entirely at random. By chance, one group may end up systematically older or predominantly one sex. Since these characteristics predict weight loss, the observed difference between diets would partly reflect this imbalance rather than the diet effect itself. Additionally, between-animal variation in baseline characteristics inflates $\text{Var}(\hat{\tau})$, reducing precision. The matched pair design removes both concerns: pairing controls for known sources of variation, and within-pair differencing eliminates between-pair variation from the variance formula.

(ii) What is the total number of valid treatment assignments $|\mathcal{W}^+|$? Compare to a CRE with $N_T = N_C = 10$.

Within each of the $P = 10$ pairs, exactly one animal receives diet A, chosen by an independent coin flip:

$$|\mathcal{W}^+| = 2^{10} = 1,024.$$

In a CRE with $N_T = N_C = 10$, all balanced assignments are valid:

$$\binom{20}{10} = 184,756.$$

The matched pair design has far fewer valid assignments because the within-pair constraint eliminates all assignments that do not place exactly one treated animal per pair.

(iii) Express the data as a table showing observed and missing potential outcomes and the assignment vector $\mathbf{w}$.

The number in parentheses shows which animal received diet A ($W_{i,p} = 1$); the other received diet B ($W_{i,p} = 0$).

Unit i	Pair p	$W_{i,p}$	$Y_{i,p}(0)$	$Y_{i,p}(1)$	$Y_{i,p}^{obs}$
1	1	1	?	13.2	13.2
2	1	0	14.0	?	14.0
1	2	0	8.8	?	8.8
2	2	1	?	8.2	8.2
1	3	1	?	10.9	10.9
2	3	0	11.2	?	11.2
1	4	1	?	14.3	14.3
2	4	0	14.2	?	14.2
1	5	0	11.8	?	11.8
2	5	1	?	10.7	10.7
1	6	0	6.4	?	6.4
2	6	1	?	6.6	6.6
1	7	1	?	9.5	9.5
2	7	0	9.8	?	9.8
1	8	0	11.3	?	11.3
2	8	1	?	10.8	10.8
1	9	0	9.3	?	9.3
2	9	1	?	8.8	8.8
1	10	1	?	13.3	13.3
2	10	0	13.6	?	13.6

(iv) Perform a Fisher randomisation test of $H_0 : Y_{i,p}(1) - Y_{i,p}(0) = 0.5$ for all (i, p) . Use the difference in means. Report the p-value and state your conclusions.

Step 1: Compute the observed test statistic.

The difference in means is $T^{obs} = \bar{Y}_A^{obs} - \bar{Y}_B^{obs}$.

Pair p	1	2	3	4	5	6	7	8	9	10	Mean
$\bar{Y}_{p,A}^{obs}$	13.2	8.2	10.9	14.3	10.7	6.6	9.5	10.8	8.8	13.3	10.63
$\bar{Y}_{p,B}^{obs}$	14.0	8.8	11.2	14.2	11.8	6.4	9.8	11.3	9.3	13.6	11.04

$$T^{obs} = 10.63 - 11.04 = -0.41.$$

Step 2: Impute missing potential outcomes under H_0 .

Under $H_0 : \tau_{i,p} = 0.5$, every unit satisfies $Y_{i,p}(1) = Y_{i,p}(0) + 0.5$. We use this to fill in the missing column for each unit:

- For a **treated** unit ($W = 1$): we observe $Y(1)$, so we impute $Y(0) = Y^{obs} - 0.5$.
- For a **control** unit ($W = 0$): we observe $Y(0)$, so we impute $Y(1) = Y^{obs} + 0.5$.

This gives us a *complete* science table under H_0 :

Unit i	Pair p	$W_{i,p}$	$Y_{i,p}(0)$	$Y_{i,p}(1)$	$Y_{i,p}^{obs}$
1	1	1	$13.2 - 0.5 = 12.7$	13.2	13.2
2	1	0	14.0	$14.0 + 0.5 = 14.5$	14.0
1	2	0	8.8	$8.8 + 0.5 = 9.3$	8.8
2	2	1	$8.2 - 0.5 = 7.7$	8.2	8.2
1	3	1	$10.9 - 0.5 = 10.4$	10.9	10.9
2	3	0	11.2	$11.2 + 0.5 = 11.7$	11.2
1	4	1	$14.3 - 0.5 = 13.8$	14.3	14.3
2	4	0	14.2	$14.2 + 0.5 = 14.7$	14.2
1	5	0	11.8	$11.8 + 0.5 = 12.3$	11.8
2	5	1	$10.7 - 0.5 = 10.2$	10.7	10.7
1	6	0	6.4	$6.4 + 0.5 = 6.9$	6.4
2	6	1	$6.6 - 0.5 = 6.1$	6.6	6.6
1	7	1	$9.5 - 0.5 = 9.0$	9.5	9.5
2	7	0	9.8	$9.8 + 0.5 = 10.3$	9.8
1	8	0	11.3	$11.3 + 0.5 = 11.8$	11.3
2	8	1	$10.8 - 0.5 = 10.3$	10.8	10.8
1	9	0	9.3	$9.3 + 0.5 = 9.8$	9.3
2	9	1	$8.8 - 0.5 = 8.3$	8.8	8.8
1	10	1	$13.3 - 0.5 = 12.8$	13.3	13.3
2	10	0	13.6	$13.6 + 0.5 = 14.1$	13.6

Step 3: Build the reference distribution.

There are $2^{10} = 1,024$ valid assignments. For each permutation, we read off $Y(1)$ for the units assigned to treatment and $Y(0)$ for the units assigned to control from the imputed table above, and compute $T = \bar{Y}(1)_{\text{treated}} - \bar{Y}(0)_{\text{control}}$.

To illustrate: for pair 1, the two possible contributions are:

- Original assignment (animal 1 treated): $Y_1(1) - Y_2(0) = 13.2 - 14.0 = -0.8$
- Flipped assignment (animal 2 treated): $Y_2(1) - Y_1(0) = 14.5 - 12.7 = 1.8$

The same logic applies in each pair independently. Since all 10 pairs favour the original assignment (flipping any pair moves T upward), the observed statistic $T^{obs} = -0.41$ is the most extreme value in the lower tail, and $+0.91$ is the most extreme in the upper tail. Only these 2 permutations out of 1,024 are at least as extreme as T^{obs} in absolute value.

Step 4: Compute the p-value.

$$p\text{-value} = \frac{\#\{|T(W)| \geq |T^{obs}|\}}{1,024} = \frac{2}{1,024} \approx 0.002.$$

Conclusion. We reject $H_0 : \tau_{i,p} = 0.5$ at the 5% level ($p \approx 0.002$). The data provide strong evidence that the treatment effect is not 0.5. Since $T^{obs} = -0.41$, diet A produces less weight loss than diet B, contrary to the null hypothesis of a positive benefit.

(v) Write down the estimator $\hat{\tau}$. Estimate τ , compute $\widehat{\text{Var}}(\hat{\tau})$, and provide a 95% confidence interval.

Estimator.

$$\hat{\tau} = \bar{Y}_A^{obs} - \bar{Y}_B^{obs} = \frac{1}{P} \sum_{p=1}^P \hat{\tau}_p, \quad \hat{\tau}_p = Y_{p,A}^{obs} - Y_{p,B}^{obs}.$$

Estimate. From the table in Part (iv): $\hat{\tau} = -4.1/10 = -0.41$.

Estimated variance.

$$\begin{aligned}\widehat{\text{Var}}(\hat{\tau}) &= \frac{1}{P(P-1)} \sum_{p=1}^P (\hat{\tau}_p - \hat{\tau})^2 = \frac{1}{90} \sum_{p=1}^{10} (\hat{\tau}_p + 0.41)^2. \\ &= \frac{0.1521 + 0.0361 + 0.0121 + 0.2601 + 0.4761 + 0.3721 + 0.0121 + 0.0081 + 0.0081 + 0.0121}{90} \\ &= \frac{1.349}{90} \approx 0.015, \quad \widehat{\text{SE}}(\hat{\tau}) = \sqrt{0.015} \approx 0.122.\end{aligned}$$

95% confidence interval. With $P - 1 = 9$ d.f., $t_{0.025,9} = 2.262$:

$$-0.41 \pm 2.262 \times 0.122 = -0.41 \pm 0.277 \implies 95\% \text{ CI} : (-0.687, -0.133).$$

The interval lies entirely below zero: diet A produces statistically significantly less weight loss than diet B at the 5% level.